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PART - A

Q1

Encapsulation:

* Encapsulation is the process of binding data and functions together into a single unit called a class.

* It also provides data hiding by restricting direct access to data using access specifiers such as private and public.

Q2

Inheritance:

* Inheritance is a feature in Object Oriented Programming in which new class aquires the properties and methods of an existing class.

* It promotes code reusability and establishes a parent - child relationship between classes.

Q3 Constructor:

A constructor is a special member function of a class used to initialize objects. It has the same name as the class, has no return type, and is called automatically whenever an object is created.

Q4 Polymorphism:

* It means "many forms". It allows the same function or interface to behave differently depending on the situation.
* The two main types of polymorphism are
  1. compile-time polymorphism
  2. Run-time polymorphism

Q5 Data Abstraction:

Data Abstraction is the process of hiding implementation details and showing only the essential features of an object to the user. It can be achieved using classes, access specifiers, and abstract interfaces.

PART - B

Q6 Inheritance is an OOP mechanism that allows a derived class to inherit properties and methods from a base class. It helps reduce code duplications and improves code reusability

Example:

```
class Animal{
   public:
      void eat(){
            cout<<"Animal can eat"; }}
class dog : public Animal{
   public:
         void bark(){
            cout<<"Dog can bark";
   }}
```

In this example, Dog is the derived class and animal is the base class. The Dog class can use the eat() function without ~~~~ redefining it.

Advantages of inheritance:

1. Code reusability
2. Reduced duplication
3. Easier maintence of programs.

Q7 Encapsulation provides data hiding and protects sensitive information from unauthorized access. Data members can be declared private and accessed only through public methods.

Advantages:

1. Improved security, direct modification of data is prevented.

2. Better maintainability, Internal implementation can be changed without affecting other parts of the program

3. Controlled Access, Getter and setter methods allow validation before data is modified.

Example:

```
class BankAccount{
    private:
        int balance;
    public:
        void setBalance (int b) {
            if(b>=0)
            balance=b; }
            int getBalance(){
                    return balance; } };
```

In this example, the balance variable cannot be modified directly outside the class which improves security and reliability.

Q8